

Computer Malware

Understanding Viruses and Other Malicious Software

"Malware" is short for **malicious software** — any program or code designed to damage, disrupt, spy on, or gain unauthorized access to a computer system. This guide is written for awareness and defense: it explains what malware is, the main categories you're likely to encounter, how infections typically happen, and how to protect yourself. It does not include any working malicious code or step-by-step instructions for creating malware.

1. What Is Malware?

Malware is an umbrella term for software created with harmful intent. Unlike a normal program bug (which is accidental), malware is deliberately designed to do something the computer's owner did not authorize — whether that's stealing data, corrupting files, spying on activity, or hijacking system resources.

Malware is generally classified by **how it spreads** and **what it does** once it's on a system. The sections below cover the most common categories.

2. Types of Malware

2.1 Viruses

A virus is malicious code that attaches itself to a legitimate program or file and spreads when that program is run or the file is opened, often infecting other files in the process. Like a biological virus, it needs a "host" to replicate and typically requires some user action (like opening an infected file) to activate.

Real-world examples: ILOVEYOU (2000), Melissa (1999), Sasser (2004).

2.2 Worms

A worm is similar to a virus but can self-replicate and spread across networks **without** needing a host file or any user action, often exploiting security vulnerabilities to jump from one machine to another automatically.

Real-world examples: WannaCry (2017), Conficker (2008), Code Red (2001).

2.3 Trojans (Trojan Horses)

A trojan disguises itself as a legitimate or useful program to trick users into installing it. Unlike viruses and worms, trojans don't self-replicate; instead they rely on deception to gain entry, often opening a "backdoor" for further attacks.

Real-world examples: Zeus, Emotet, Trojan.Win32.Agent variants.

2.4 Ransomware

Ransomware encrypts a victim's files or locks them out of their system, then demands payment (usually in cryptocurrency) in exchange for restoring access. It has become one of the most financially damaging categories of malware for both individuals and organizations.

Real-world examples: WannaCry, Ryuk, LockBit, CryptoLocker.

2.5 Spyware

Spyware secretly monitors a user's activity — browsing habits, keystrokes, credentials, or personal files — and sends that information to a third party, usually without the victim's knowledge.

Real-world examples: Pegasus (mobile spyware), FinFisher, CoolWebSearch.

2.6 Keyloggers

A specific type of spyware that records every keystroke a user types, often used to steal passwords, credit card numbers, and other sensitive information.

Real-world examples: Ardamax Keylogger, HawkEye (as commonly discussed in security reports).

2.7 Adware

Adware automatically displays or downloads unwanted advertising, often bundled with free software. While frequently more annoying than dangerous, some adware also tracks user behavior or opens the door to more serious infections.

Real-world examples: Fireball, Appearch, DollarRevenue.

2.8 Rootkits

A rootkit is designed to gain privileged (administrator/root-level) access to a system while hiding its own presence and that of other malware, making it extremely difficult to detect and remove using normal tools.

Real-world examples: Sony BMG rootkit scandal (2005), Zacinlo, TDSS/TDL-4.

2.9 Botnets & Bots

A "bot" is a compromised computer secretly controlled remotely by an attacker; a "botnet" is a large network of such machines, often used together to send spam, launch distributed denial-of-service (DDoS) attacks, or mine cryptocurrency.

Real-world examples: Mirai, Emotet botnet, 3ve.

2.10 Fileless Malware

Rather than installing a file on disk, fileless malware operates directly in a computer's memory (RAM) or abuses legitimate system tools, making it harder for traditional antivirus software to detect.

Real-world examples: Astaroth, Frodo (as historically documented), PowerShell-based attacks.

2.11 Logic Bombs

Malicious code that lies dormant until triggered by a specific condition, such as a certain date or a particular user action, at which point it executes its harmful payload.

Real-world examples: Often embedded within other malware or by insider threats rather than distributed standalone.

3. Common Infection Methods

Malware typically reaches a system through one of the following channels:

- **Phishing emails** — malicious attachments or links disguised as legitimate messages
 - **Malicious or compromised websites** — drive-by downloads that install malware just by visiting a page
 - **Infected removable media** — USB drives or external disks carrying hidden malware
 - **Software vulnerabilities** — exploiting unpatched security flaws in operating systems or applications
 - **Bundled software** — malware hidden inside "free" downloads, cracked software, or fake installers
 - **Social engineering** — tricking users into disabling security features or granting permissions
-

4. Common Signs of a Malware Infection

- Unusually slow performance or frequent crashes
- Unexpected pop-ups, new toolbars, or a changed browser homepage
- Programs opening, closing, or installing themselves without permission
- Unexplained data usage, disk activity, or battery drain
- Files becoming encrypted, missing, or renamed unexpectedly
- Antivirus software being disabled without your action
- Friends or contacts receiving strange messages from your accounts

5. How to Protect Against Malware

- 1 **Keep software updated.** Apply operating system and application patches promptly to close known security holes.
 - 2 **Use reputable antivirus/anti-malware software.** Keep it running and its definitions up to date.
 - 3 **Enable a firewall.** Use your OS's built-in firewall or a dedicated one to filter suspicious network traffic.
 - 4 **Be cautious with email and links.** Avoid opening attachments or clicking links from unknown or unexpected senders.
 - 5 **Download software only from official sources.** Avoid pirated software and untrusted third-party download sites.
 - 6 **Back up important data regularly.** Keep offline or cloud backups so ransomware or data loss can't wipe out everything.
 - 7 **Use strong, unique passwords and multi-factor authentication.** This limits the damage if credentials are ever stolen.
 - 8 **Limit user privileges.** Avoid running everyday tasks from an administrator account when a standard account will do.
-

6. Quick Reference Table

Type	Spreads By	Primary Goal
Virus	Attaching to files/programs	Damage or corrupt data
Worm	Self-replicating over networks	Spread rapidly, consume resources
Trojan	Disguised as legitimate software	Gain unauthorized access
Ransomware	Various (email, exploits)	Extort payment via encryption
Spyware	Bundled installs, exploits	Covertly gather information
Adware	Bundled with free software	Display unwanted ads
Rootkit	Exploits, trojans	Hide presence, maintain access
Botnet malware	Worms, trojans, exploits	Remote control for coordinated attacks

7. Summary

Malware covers a wide range of malicious software — from self-replicating viruses and worms to deceptive trojans, extortion-driven ransomware, and covert spyware. Most infections rely on some form of deception or an unpatched vulnerability to get a foothold. The best defense combines up-to-date

software, reliable security tools, cautious online habits, and regular backups, so that even if an infection occurs, the damage can be contained and reversed.